GreenEquip: Decentralized Farm Machinery Rental Application using Blockchain

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***Abstract*—For small and medium-sized farmers, access to reasonably priced agricultural equipment continues to be a significant obstacle because buying machinery is costly and traditional rental methods frequently lack transparency and trust. By creating GreenEquip, a blockchain-based platform that permits peer-to-peer farming equipment rentals without the need for middlemen, this project tackles the issue. Farmers and equipment owners may simply interact with the platform through a web interface connected via MetaMask, while the system employs an Ethereum smart contract to manage equipment listings, bookings, and payments. A secure and useful method of sharing agricultural resources was demonstrated by the application's testing, which revealed that equipment could be successfully listed, rented, and tracked in real time with each transaction being permanently recorded on the blockchain.**

***Index Terms*—Transparency, blockchain, peer-to-peer, MetaMask.**

I. Introduction

Smallholder farmers continue to encounter obstacles in obtaining contemporary equipment through conventional leasing systems that are expensive, opaque, and reliant on middlemen, despite the agricultural sector's continued importance to global livelihoods. These challenges reduce output, deter the adoption of better farming techniques, and exacerbate persistent economic fragility. GreenEquip was developed to solve these problems by providing a dependable and transparent means of direct communication between farmers and equipment owners. The platform promotes responsible resource use through a shared economy model that lowers underutilization and increases affordability, which is in line with Sustainable Development Goal 12.

The technology creates a safe peer-to-peer marketplace where all transactions are recorded immutably and carried out without the need for centralized control by utilizing the Ethereum blockchain and smart contracts. With the help of a simple web interface created with HTML, Tailwind CSS, JavaScript, and Ethers.js and a Solidity-based backend, customers may register equipment, peruse listings, and rent machines using their MetaMask wallets. By doing this, GreenEquip shows how blockchain technology and contemporary web development may work together to address a significant agricultural problem while highlighting useful uses of decentralized technology.

II. Literature survey

Blockchain, decentralized data systems, and digital platforms are becoming increasingly important in modern agriculture, according to a large body of recent study. Important issues like lack of transparency, reliance on middlemen, expensive equipment, and ineffective resource allocation are frequently mentioned in the literature. In response, numerous researches suggest decentralized, verifiable solutions meant to assist farmers by enhancing resource accessibility, safeguarding data sharing, and streamlining operational procedures. When taken as a whole, these pieces offer a strong conceptual basis for a decentralized platform like GreenEquip that rents out agricultural equipment.

Numerous studies highlight the significance of safe information management and traceability. Blockchain can be used to track agricultural products throughout their whole lifecycle, guaranteeing that all pertinent data stays tamper-proof and transparently available to stakeholders, according to research published in [1]. By eliminating middlemen and connecting IoT-generated data to unchangeable ledgers, the approach seeks to increase trust. The fundamental ideas underpin GreenEquip's strategy for keeping clear and lasting records of equipment ownership and rental activities, notwithstanding the company's emphasis on product provenance rather than machinery. In a similar vein, [12] suggests a distributed data ecosystem that addresses issues with centralized IoT-based traceability solutions, including limited auditability and susceptibility to data tampering. This supports the claim that decentralization enhances agricultural information systems' dependability and integrity.

The security and preservation of agricultural assets are the subject of additional contributions. Blockchain can protect digital

asset information in dispersed situations, as demonstrated by the methodology outlined in [2]. The study offers a justification for using blockchain to handle equipment registrations, rental agreements, and financial transactions by viewing farming resources as assets that need safe verification and recordkeeping. In addition, [9] explores the technological underpinnings of blockchain, including cryptographic tools, smart contracts, and consensus processes, and looks at how these may be used in a variety of fields, including contract automation, provenance tracking, food safety, and smart farming data. The study supports the idea that long-standing problems with fraud, manipulation, and ineffective middlemen in agricultural systems can be resolved by blockchain's trust paradigm.

The challenges farmers encounter in obtaining agricultural equipment are covered in a large amount of the literature. [3] presents a concept that uses smart contracts to automate the loan and renting process, doing away with the necessity for human participation and lowering the possibility of disputes. The approach strongly correlates with GreenEquip's focus on dependable, rule-based rental workflows by creating verifiable agreements and automatic execution. Conventional digital solutions like [5] and [10] draw attention to the practical difficulties associated with equipment access, such as variable rental costs, lengthy travel times, and restricted availability. These pieces highlight the necessity of platforms that offer consistency, accessibility, and clarity. Their findings highlight the usefulness of solutions such as GreenEquip, which expand these advantages through immutable recordkeeping and decentralization.

More advanced strategies use intelligence-driven decision models and go beyond straightforward rental logistics. [11] suggests an algorithmic engine that can analyze factors like crop variety, harvest time, and farmer budgets to produce customized equipment suggestions. This strategy emphasizes how important structured decision assistance is to ensuring small-scale farmers have timely and affordable access. By looking at cooperative sharing models that encourage cooperative resource consumption, lower total costs, and promote sustainability, the study titled [6] provides another dimension. The conceptual underpinnings of GreenEquip's peer-to-peer rental architecture are reinforced by such work, which highlights the financial and ecological advantages of shared equipment systems.

The coordination and scheduling of machinery on a community-wide scale is addressed in [8]. This study suggests using an improved evolutionary algorithm in conjunction with a consortium blockchain to effectively distribute idle equipment among a user network. While digital currency-based payments simplify transactions, its crowdsourcing-based scheduling system reduces resource waste. Consensus efficiency and system stability are improved by adding trusted supervisory nodes to validate blocks. These results show that decentralized coordination methods are feasible, which is theoretically consistent with GreenEquip's goal of optimizing equipment distribution through transparent, on-chain procedures.

The literature consistently highlights blockchain as a practical solution for improving transparency, trust, and efficiency in agricultural systems.

III. Methodology

It is crucial to comprehend a number of fundamental ideas that underlie blockchain technology and its use in this decentralized rental marketplace before diving into GreenEquip's technical implementation.

Blockchain and Distributed Ledgers:

At its core, a blockchain is a distributed ledger. Instead of being kept on a single centralized server, it is a database that is replicated among several computers (nodes) in a network. A set of transactions is contained in each block of the chain, and once a block is added to the chain, it is essentially unchangeable. This results in a permanent and verifiable audit trail since historical records cannot be changed or removed without being discovered. Every equipment registration, rental agreement, and payment transaction in the GreenEquip context is documented on the Ethereum blockchain, where it can be independently confirmed by any network participant.

Smart Contracts:

Self-executing agreements with predetermined rules that are directly inscribed onto the blockchain are known as smart contracts. Smart contracts automatically carry out their logic when predefined criteria are satisfied, as contrast to traditional contracts that need middlemen like attorneys or notaries to understand and enforce provisions. When a farmer wants to rent a tractor through GreenEquip, for example, the smart contract automatically updates the equipment status, transfers the payment from the renter's wallet to the owner's wallet, locks the rental agreement, and confirms that the equipment is available; all without the need for human intervention. In addition to cutting expenses, this automation removes the potential for disagreements resulting from incorrect interpretation or selective implementation of contract provisions.

Decentralization and Peer-to-Peer Networks:

Conventional online markets function by use of centralized platforms that serve as middlemen between consumers and sellers. On the other hand, equipment owners and renters communicate directly with one another over the blockchain in a decentralized application like GreenEquip, which functions on a peer-to-peer basis. There isn't a single entity that can censor users, tamper with transactions, or take money from the ecosystem. In agricultural communities, where trust in centralized institutions may be low and the removal of intermediate fees can have a substantial impact on the economic feasibility of equipment sharing, this design is especially beneficial.

Cryptocurrency and Ether (ETH):

To enable transactions, blockchain networks need a local digital currency. This money is known as Ether (ETH) on the Ethereum blockchain. When a farmer uses GreenEquip to rent equipment, the owner's wallet receives the ETH rental payment from the renter's digital wallet. Through wallet apps like MetaMask, which offer a safe interface for handling private keys and signing transactions, users engage with their ETH holdings.

Figure 1 describes how the GreenEquip system is structured around three layers, each layer serving its purpose:

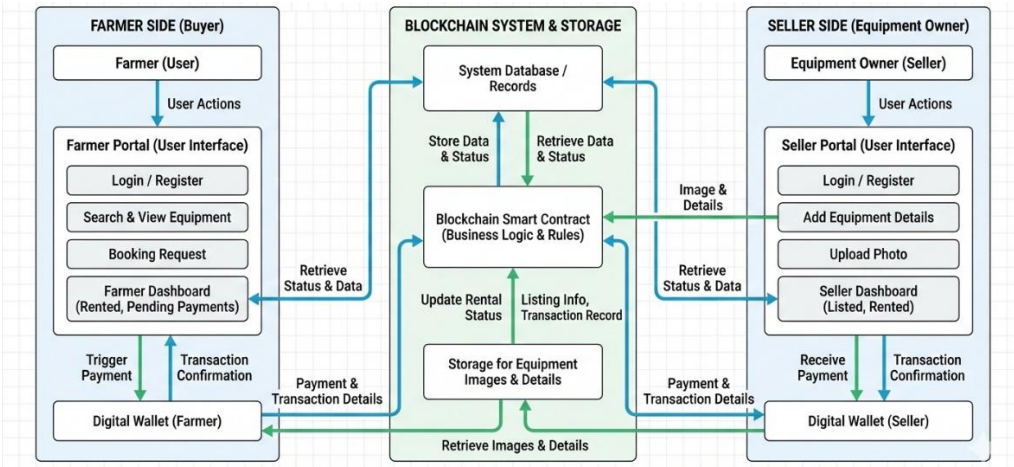

Fig. 1. Block Diagram of the System

A. Farmer Portal Layer (Buyer)

The main interface for users looking to rent agricultural equipment is the Farmer Portal layer. All farmer-side interactions, such as user authentication, equipment finding, booking initiation, and rental monitoring, are handled by this layer. Farmers can access equipment photos that are retrieved from the underlying blockchain-backed storage system, view rental rates, and per use available equipment listings.

The site calculates the rental rate depending on the chosen length when a farmer chooses an equipment piece and submits a booking request. The following formula is used to get the total rental amount:

$$\text{Total Rental Cost} = R \times D \tag{1}$$

Equation (1) shows the formula for total rental cost, where D stands for the number of rental days and R is the equipment's daily rental fee. To ensure pricing transparency, this computed value is given to the farmer before payment is confirmed.

The Farmer Portal initiates a payment request via the farmer's digital wallet after confirmation. The seller and the gateway do not currently communicate directly. Rather, the request is sent to the blockchain system for execution and verification. Users can track rented equipment and outstanding payments in real time thanks to the farmer dashboard, which continuously retrieves transaction confirmations and rental status updates from the blockchain.

B. Seller Portal Layer (Equipment Owner)

The Seller Portal layer allows equipment owners to list, manage, and monitor their agricultural assets. Through this website, sellers can register or identify themselves and input equipment specifics, such as rental rate, availability status, and descriptive information. Through the website, equipment images are provided and linked to pertinent system data.

Once equipment is listed, farmers on the site can locate it. The Seller Portal provides a dashboard that shows the current status of each advertised item, including whether it is available, rented, or awaiting confirmation. The rental updates and payment notifications on this dashboard are derived directly from blockchain records, ensuring accuracy and preventing unauthorized modifications. After a rental transaction is completed, the payment is transmitted directly to the seller's digital wallet via the blockchain system. This eliminates the need for manual payment reconciliation and ensures that vendors only get paid if the rental criteria are fulfilled.

C. Blockchain System and Storage Layer

GreenEquip's basic execution environment is made up of the Blockchain System and Storage layer. Enforcing rental regulations, verifying transactions, and keeping unchangeable records of all system activity fall under the purview of this layer. This layer handles all booking requests made by farmers and listing updates supplied by sellers.

The blockchain system confirms the availability of equipment, the length of the rental, and the authenticity of the payment upon receiving a booking request. The system only completes the transaction—transferring the rental payment to the seller and changing the equipment status to "rented"—after a successful verification. Because these operations are carried out atomically, partial or inconsistent states cannot arise.

To ensure transparency and traceability, transaction records, rental status updates, and listing details are kept on the blockchain. Images of the equipment and descriptive information are kept in the storage component and are retrieved by either portal as needed. The system guarantees that neither party may alter rental results or payment records by centralizing trust enforcement within this layer.

In Figure 2, the interaction flow between the equipment renter, equipment owner, GreenEquip application, and the blockchain network is illustrated. The diagram captures the end-to-end communication involved in equipment listing, rental initiation, transaction execution, and confirmation, highlighting how blockchain facilitates secure and transparent coordination among system participants.

The workflow begins with the Equipment Owner, who submits equipment-related details to the GreenEquip application. These details include equipment information and the owner's wallet signature, which serves as cryptographic proof of ownership and intent. Upon receiving this information, the GreenEquip application forwards the registration data to the blockchain layer, where the equipment listing is securely recorded. Once the registration process is completed, confirmation messages and future rental notifications are communicated back to the equipment owner through the application interface.

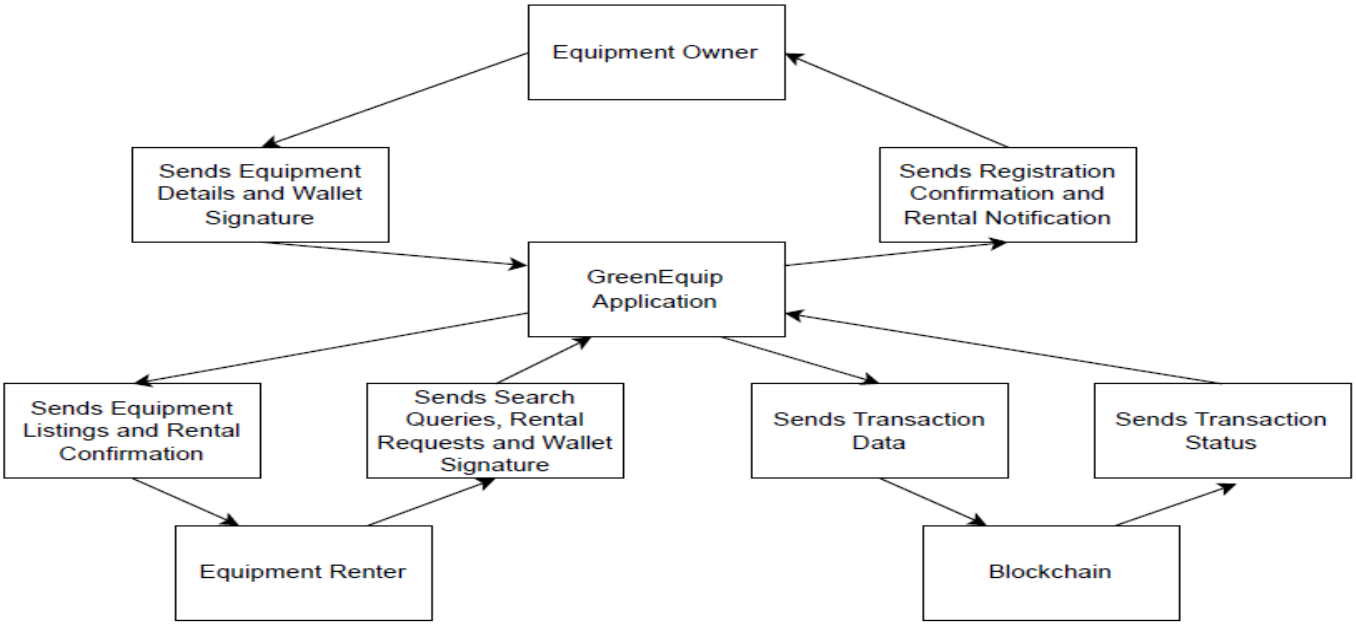

Fig. 2. Diagram representing relationship between owner, renter, and use of blockchain technology

The Equipment Owner starts the process by entering information about the equipment into the GreenEquip program. The owner's wallet signature, which acts as cryptographic evidence of ownership and purpose, and equipment details are among these details. After obtaining this data, the GreenEquip application transmits the registration information to the blockchain layer, where the equipment listing is safely stored. The equipment owner receives confirmation messages and future rental notifications via the application interface after the registration process is finished.

The Equipment Renter uses the GreenEquip application to send search queries and rental requests in order to communicate with the system. The renter is shown listings of available equipment that are retrieved from blockchain data. The renter offers a wallet signature to approve the transaction when they choose an equipment item and move on with a rental request. The renter's unequivocal approval of the rental terms and related payment is ensured by this signature.

By gathering signed transaction data from both parties and sending it to the blockchain network, the GreenEquip application serves as an intermediary coordination layer. By authenticating payment data, monitoring equipment availability, and validating signatures, the blockchain verifies the transaction. The rental agreement is permanently entered into the distributed ledger after the transaction has been confirmed.

The GreenEquip application receives transaction status notifications from the blockchain following successful execution. These updates contain the equipment owner's payment confirmation and the renter's rental confirmation. In order to guarantee that both parties are promptly and consistently notified of the transaction outcome, the application then transmits this information to the corresponding users.

The technology removes the requirement for direct trust between equipment owners and renters by passing all crucial processes through the blockchain. Rental agreements and payments are safely enforced because to the unchangeable transaction logs and cryptographic wallet signatures. This interaction paradigm creates a dependable peer-to-peer leasing mechanism appropriate for decentralized agricultural equipment sharing, improves transparency, and lessens disagreements.

IV. Results

The GreenEquip decentralized application was deployed and tested on the Ethereum blockchain to evaluate its functional correctness, transparency, and user interaction workflow. The results demonstrate that the system successfully enables trustless peer-to-peer rental of agricultural equipment without reliance on centralized intermediaries.

Upon equipment registration by the seller, each asset was assigned a unique on-chain identifier and stored within the smart contract state. Associated metadata, including equipment images and descriptive details, were uploaded to decentralized storage and linked to the blockchain using cryptographic content hashes. This ensured tamper-resistant storage while maintaining scalability and reduced on-chain data costs. Registered equipment became immediately visible to potential renters through the farmer portal.

When a farmer selected an equipment item and initiated a booking, the system prompted a MetaMask-based payment transaction. Successful payment execution resulted in the automatic generation of a blockchain transaction hash, which served as verifiable proof of the rental agreement. The booking state was updated in real time from payment pending to confirmed, and the rented equipment was dynamically removed from the available listings.

The renter dashboard reflected all active and completed rentals, clearly distinguishing between confirmed bookings and those awaiting payment approval. Simultaneously, the seller portal displayed an updated list of listed equipment, highlighting items currently rented and preventing duplicate bookings. These state transitions were enforced entirely by the smart contract, eliminating the need for manual verification.

The rental interaction produced a reciept for the buyer, where the details of the booking are written down, as shown in Fig. 3:

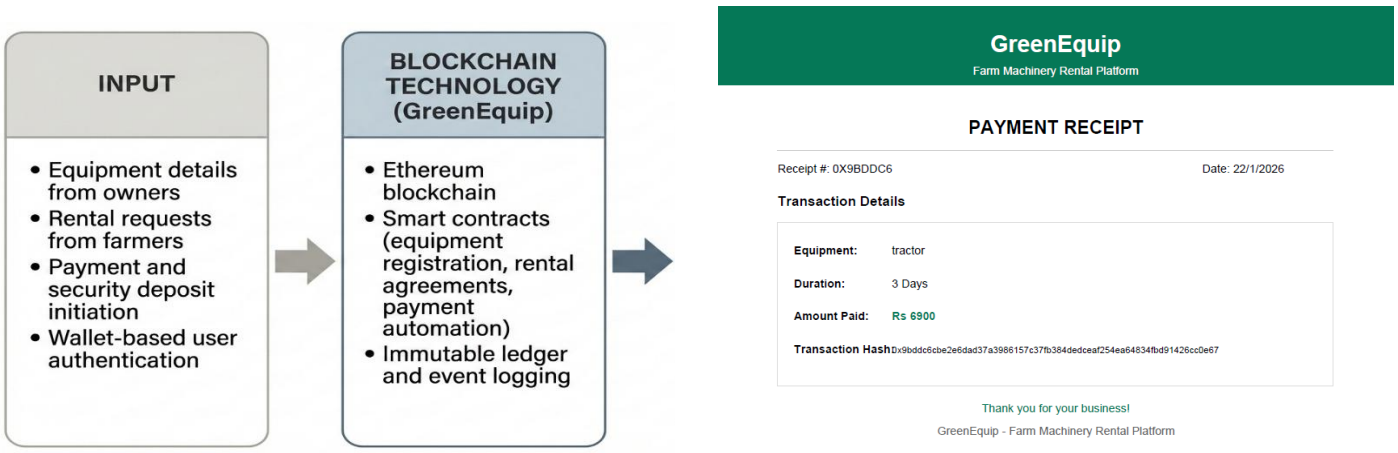

Fig. 3. Diagram representing relationship between owner, renter, and use of blockchain technology

Each rental interaction produced an immutable on-chain record containing the renter address, rental duration, payment amount, and equipment identifier. These records were publicly verifiable using Ethereum block explorers, demonstrating full transparency and traceability of transactions. Additionally, the system successfully handled failed or cancelled transactions by reverting state changes and preserving data consistency.

Performance evaluation showed that equipment registration and booking operations were completed within a few seconds, depending on network congestion, with predictable gas consumption. The results confirm that GreenEquip achieves secure asset listing, automated payment handling, and real-time rental state management using decentralized technologies, validating its feasibility as a blockchain-based agricultural rental platform.

V. Conclusion

The GreenEquip system effectively illustrates how blockchain technology may be used to construct a decentralized agricultural equipment rental business. Through smart contracts, the developed system facilitates safe equipment listing, rental execution, and payment settlement, guaranteeing that all transactions are publicly documented and automatically enforced without human interference. The findings demonstrate that rental activities can be carried out dependably while preserving transaction consistency and data integrity.

Through the implemented three-layer architecture, the system effectively separates user interaction from transaction enforcement. Farmers were able to browse equipment, calculate rental costs, and complete bookings with immediate confirmation, while equipment owners could list assets and receive payments directly upon successful rental execution. Blockchain-based transaction handling ensured that equipment availability, rental status updates, and payment transfers were synchronized across both user portals in real time.

The experimental process verified that using immutable ledger records prevented frequent problems like double booking and payment conflicts and decreased ambiguity in rental agreements. Every rental that was finished produced a verified transaction record that gave both parties accountability and traceability. These results demonstrate the system's capacity to function as a peer-to-peer, trustless marketplace.

Overall, the findings confirm that implementing blockchain-powered rental systems in agricultural settings is feasible. GreenEquip provides a scalable solution for increasing access to vital farming resources by demonstrating how decentralized automation may improve efficiency, transparency, and dependability in equipment sharing.

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